

The dual intertwiner

Moment Gram, Borodin dual, and the fractional-linear dictionary

$$R = C(I + C)^{-1}$$

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Abstract

Rounds 404 and 405 stopped at the same named remainder: Euler/Digamma Grams are exact on the weight layer, while $A_0 = R_{N-3} - \frac{1}{2}I$ lives behind the dual resolvent. This note writes the exact intertwiner. Let $u^\vee$ be the Borodin dual of $|u|$ and let C be the Christoffel–Darboux Gram of the μ -only dual chain, evaluated on the hole nodes Y and dressed by $\sqrt{u^\vee|_Y}$. Then

$$R_{N-3} = C(I + C)^{-1},$$

with $\text{eig}(R) = \lambda(C)/(1 + \lambda(C))$, hence $\text{ind}_- A_0 = \#\{\lambda(C) < 1\}$ and $\text{P1} \iff \lambda_2(C) \geq 1$. The ν -Hankel after compression has rank N_ν , not one. The Loewner sandwich of the inverse upper-bounds R and cannot force $R \succeq \frac{1}{2}I$. The Euler source Gram is class SOURCE_GRAM_EXACT on the moment layer of u , which is *not* the Gram that R inverts. No RH claim.

Status. Lemmas 1–4 are proved (finite identities over $\mathbb{Q}$). Lemmas 5–6 and Propositions 2–4 are a finite machine census on FRAME-A, core-42, and the χ_3 twins. No RH claim. No L^* claim. No $R^\dagger \succ \frac{1}{2}I$ claim.

1 The formula

Write u for the folded signed union weight on the cosine grid X , $Y = \text{supp}(u_-)$ the hole nodes, and $u_j^\vee = (1 - x_j)^2 / (S|u_j|\kappa_j^2)$ the Borodin dual (r356). Let $n = N_w - 3$. In the Chebyshev basis $(T_j)_{j < n}$ let G_+ be the Hankel Gram of $u^\vee$ on the μ -nodes, Φ the Chebyshev–Vandermonde on Y , $D = \text{diag}(u^\vee|_Y)$, and $A = D^{1/2}\Phi$. Then $G_\nu = A^T A$ and $G = G_+ + G_\nu$, and the dual CD Gram of depth n on Y is the congruence

$$R_{N-3} = AG^{-1}A^T = C(I + C)^{-1}, \quad C := AG_+^{-1}A^T. \quad (1)$$

The second form is Woodbury: $A(G_+ + A^T A)^{-1}A^T = C(I + C)^{-1}$. The three-term recurrence of $u^\vee|_\mu$ is the Cholesky of G_+ in the OP basis, so $C = BB^T$ with B the μ -OP matrix evaluated on Y and dressed by $\sqrt{u^\vee|_Y}$ — the numerically stable form of (1) at working depth (Hankel $\kappa(G) \sim 10^{11}$ on a deep razor window mis-reads $\lambda_2(C) - 1$).

Lemma 1 (Hankel linearity). *Over $\mathbb{Q}$, the monomial Hankel satisfies $G(\mu - \nu) = G(\mu) - G(\nu)$ and $G(\mu) = G(\mu_{p_2}) + G(\mu_{p_3})$ on a five-atom toy.*

Lemma 2 (Woodbury and $R = C(I + C)^{-1}$). *Over $\mathbb{Q}$: the 2×2 Woodbury identity with $\eta = \frac{5}{6}$, $\det = 11$; and for $C = \text{diag}(1, 2)$,*

$$C(I + C)^{-1} = I - (I + C)^{-1} = \text{diag}\left(\frac{1}{2}, \frac{2}{3}\right).$$

Lemma 3 (Inverse is Loewner-antitone). *If $A = \text{diag}(2, 4)$ and $B = \text{diag}(1, 3)$ then $A \succeq B \succ 0$ and $A^{-1} \preceq B^{-1}$ over $\mathbb{Q}$.*

Lemma 4 (Fractional-linear dictionary). *If $C = \text{diag}(2, \frac{1}{2})$ then $R = C(I + C)^{-1} = \text{diag}(\frac{2}{3}, \frac{1}{3})$, so $\#\{\lambda(C) < 1\} = 1 = \text{ind}_-(R - \frac{1}{2}I)$. In exact arithmetic, $\lambda \mapsto \lambda/(1 + \lambda)$ sends $\{\lambda < 1\}$ onto $\{r < \frac{1}{2}\}$ on $[0, \infty)$.*

Lemma 5 (Machine formula). *On FRAME-A, $\|C(I + C)^{-1} - R_{\text{chain}}\|_F / \|R\|_F = 1.84 \cdot 10^{-14}$ at depth $n = 181$, and the Chebyshev–Hankel form of (1) matches the chain at depth 20 to $3.1 \cdot 10^{-15}$. The sorted fractional-linear image of $\text{eig}(C)$ matches $\text{eig}(R)$ to $1.5 \cdot 10^{-13}$. $C_{\min} = 0.85712$, $\lambda_2(C) = 1.00018$, $C_{\min}/(1 + C_{\min}) = 0.46153 = r_{\min}(R)$, $\text{ind}_- A_0 = 1$.*

Lemma 6 (Signed consumption). *Passing the signed u to the dual weight produces 104 negative dual masses on FRAME-A ($= |Y|$); the construction uses $|u|$. The source Chebyshev–Hankel of u and the dual Hankel of $u^\vee$ differ by relative residual 141 at degree 12 — R inverts the dual Gram, not the Euler-source Gram.*

2 Moment-layer Euler Gram

Proposition 1 (SOURCE_GRAM_EXACT on the moment layer). *The Chebyshev–Hankel of the folded prime-positive weights equals FF^T with features $F_{\alpha j} = T_j(x_\alpha)\sqrt{w_\alpha}$, residual $1.55 \cdot 10^{-16}$, $\lambda_{\min} > 0$. This is the source-exact Gram sought by r404, on the layer where Fold/Tent is linear (lag-residual $3.54 \cdot 10^{-16}$). A permutation of those weights yields relative residual 0.54. Node jitter of $G(\nu)$ at 10^{-3} yields residual 0.014.*

The class is tautological as a Gram of monomials, and *not* a Cholesky of A_0 . It does not transport through $u \mapsto 1/|u|$.

3 Rank, sandwich, census

Proposition 2 (ν -rank is N_ν , not one). *On FRAME-A, $\text{rank } G_\nu = 104 = |Y|$ and $\text{rank}(R - C) = 104$ ($r_{\text{eff}} = 1.18$). The dual CD tail $R_{N_w-1} - R_{N_w-3}$ is rank two (r363) and does not isolate the A_0 -defect. The one defect of MAIN is spectral: exactly one eigenvalue of C lies below 1.*

Proposition 3 (Sandwich is the wrong direction). *$C - R = C(I + C)^{-1}C \succeq 0$, so $C \succeq R$. On FRAME-A, $\lambda_{\min}(C) = 0.857 > \frac{1}{2}$ while $\lambda_{\min}(R) = 0.461 < \frac{1}{2}$. Operator monotonicity of $G \mapsto G^{-1}$ therefore upper-bounds R and cannot force $R \succeq \frac{1}{2}I$.*

Proposition 4 (Census and kills). *On core-42, $\text{ind}_- A_0 = \#\{\lambda(C) < 1\}$ on 42/42 (28 P1 and 14 vacuous). Living χ_3 : 37/37; dead χ_3 : 5/5 (the dead fulfill the transport; they die at $\text{sch} > 0$, r401). Scramble: 21 = 21. Weight-permutation: 20 = 20 (kills P1; the dictionary still holds). Wrong depth $n + 1$: relative residual 0.031, rank one.*

4 What this reduces, and what it does not

P1 on a canonical window is exactly $\lambda_2(C) \geq 1$, with C the μ -only dual CD kernel on Y . That is a threshold form on the *right* layer (the dual moment inverse), which r400 did not have. It is not a source-explicit factorisation of A_0 by Euler features: those live on $G(u)$, and $R = f(G(u^\vee))$. The Aronszajn parallel-sum of reproducing kernels is the coefficient-layer identity $(G_+ + G_\nu)^{-1} = G_+^{-1} : G_\nu^{-1}$ when both are invertible; G_ν has rank N_ν , so the update is rank- N_ν , not rank one.

Machine check

Standalone: Hankel linearity, Euler split, Woodbury, $R = C(I + C)^{-1}$, inverse antitone, FL dictionary, all over $\mathbb{Q}$. Construction: FRAME-A formula and C -anatomy, Euler moment Gram versus dual Gram, ν -rank and sandwich, scramble / permute / dead χ , core sample including the deep razor $k_z = 55$. Discovery census: `dual_intertwiner_probe.py` (`--smoke` 26/26 / full 29/29), SPEC_SHA `2ee74c59b57ed1c6`, wall 2.4s. Exit line: DUAL INTERTWINER VERIFIED.

Claim boundary

Finite identities for the Hankel, Woodbury, the fractional-linear dictionary, and a named reduction of the finite inertia of $R_{N-3} - \frac{1}{2}I$ to the inertia of $C - I$ on tested windows. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

- [1] S. Hamann, *Borodin dual hole*, sealed probe, August 2026.
- [2] S. Hamann, *Source one-defect Gram*, standalone note, August 2026, `rh/problem/one_defect_gram.tex`.
- [3] S. Hamann, *Edge contractive lift*, standalone note, August 2026, `rh/problem/edge_contractive_lift.tex`.
- [4] S. Hamann, *One-defect finite algebra*, `rh/lean/RH/OneDefect.lean`, August 2026.